

Lab - Escalado y Balanceo de Carga de una Arquitectura

Dificultad:  Intermedio

Tiempo Estimado:  45 minutos

Servicios Principales:  Amazon EC2,  Elastic Load Balancing (ELB),  Amazon EC2 Auto Scaling,  Amazon CloudWatch.

Resumen y Objetivos

En este laboratorio utilizarás Elastic Load Balancing (ELB) y Amazon EC2 Auto Scaling para balancear la carga y escalar automáticamente tu infraestructura. ELB distribuye automáticamente el tráfico de red de las aplicaciones a través de múltiples instancias EC2, mientras que Auto Scaling ayuda a mantener la disponibilidad ajustando automáticamente la capacidad según las condiciones que definas.

Objetivos:

-  Crear una AMI a partir de una instancia EC2 existente.
 -  Crear un balanceador de carga de aplicación (ALB).
 -  Crear una plantilla de lanzamiento y un grupo de Auto Scaling.
 -  Configurar el grupo de Auto Scaling para desplegar instancias en subredes privadas.
 -  Utilizar alarmas de Amazon CloudWatch para monitorear el rendimiento.
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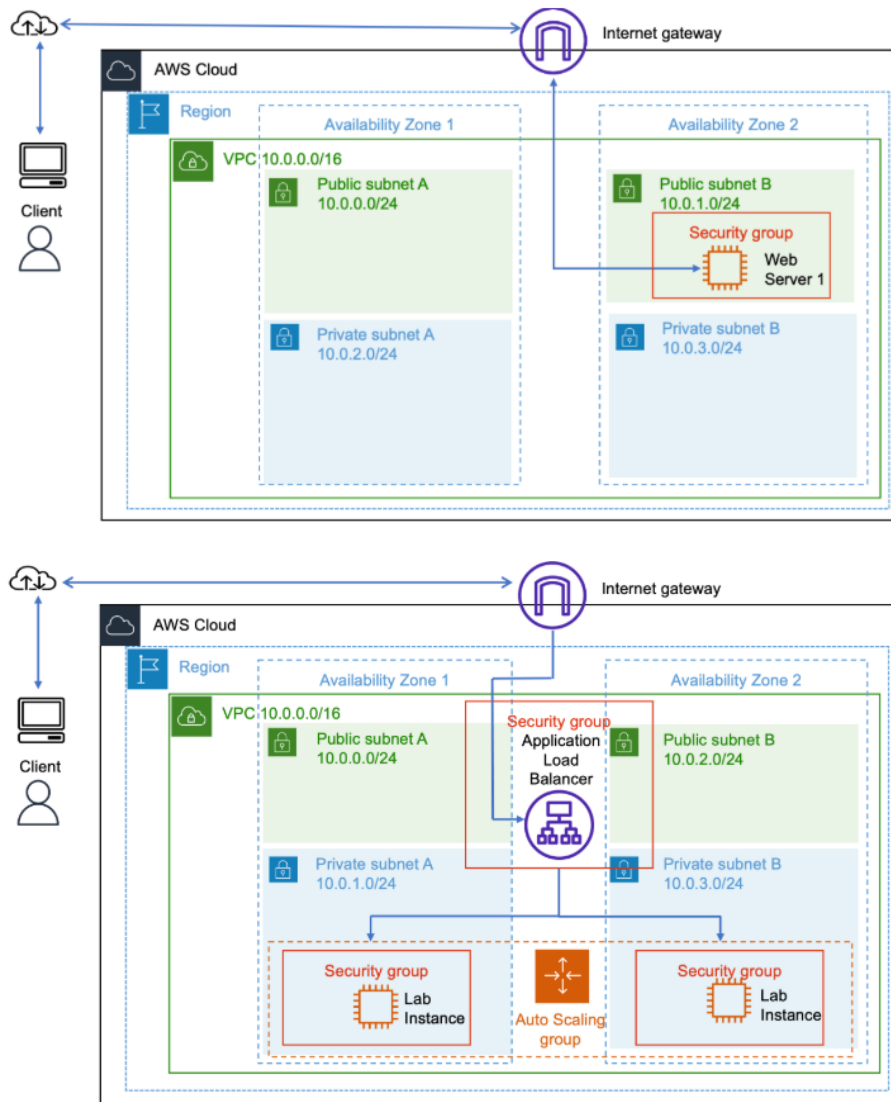
Análisis del Escenario

El cliente cuenta con una arquitectura de inicio con un servidor web simple (Web Server 1) operando en una subred pública, lo cual expone la instancia a internet directamente y representa un punto único de falla (SPOF) en caso de sufrir una sobrecarga o caída.

Se requiere implementar una arquitectura resiliente y elástica migrando dicho entorno hacia instancias aprovisionadas en subredes privadas para mayor seguridad, distribuidas dinámicamente en múltiples Zonas de Disponibilidad, todo orquestado por un Balanceador de Carga Application Load Balancer que servirá como punto único de entrada. Además, deberás implementar políticas de de Auto Scaling según patrones de utilización de la CPU (Target Tracking al 50%).

Arquitectura

-  **Arquitectura Inicial:** Infraestructura tradicional con un único "Web Server 1" expuesto dentro de una subred pública en una única AZ.
-  **Arquitectura Final:** Application Load Balancer recibiendo tráfico de internet, y enrutándolo hacia múltiples instancias de EC2 desplegadas dinámicamente (Auto Scaling) a través de subredes privadas, repartidas en 2 Zonas de Disponibilidad.



Desarrollo

🕒 Tarea 1: Creación de una AMI para Auto Scaling

En este paso, crea una Imagen de Máquina de Amazon (AMI) conservando la configuración del disco de arranque original del Web Server 1.

1. 🌐 En la consola de administración de AWS, ubica la barra de búsqueda superior, escribe **EC2** y selecciona **EC2** para abrir la consola.
2. 📁 En el panel de navegación izquierdo, localiza la sección **Instancias** y haz clic en **Instancias**.
3. ✅ Selecciona la instancia **Web Server 1** (asegúrate de que figura con el estado de *En ejecución* o *Running*).
4. ⚙️ Desde el menú desplegable superior **Acciones**, elige **Imágenes y plantillas** > **Crear imagen**.

The screenshot shows the AWS Management Console for the 'us-west-2' region. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, and Capacity Manager. The main content area displays the details for the EC2 instance 'Web Server 1' (Instance ID: i-0d66ec9e49ac0c823). The instance is in a 'Running' state, using the 't3.micro' instance type, and is located in the 'us-west-2b' Availability Zone. The instance summary includes the Instance ID, Public IPv4 address (16.145.86.117), Instance state (Running), Private IP DNS name (ip-10-0-2-137.us-west-2.compute.internal), Instance type (t3.micro), VPC ID (vpc-059c473d448ba8afe), Subnet ID (subnet-080a86c4dfb3ce566), and Instance ARN. The instance is associated with the 'IMDSv2' IAM role and the 'Public' Elastic IP address.

5. Configura los siguientes parámetros:
 - **Nombre de la imagen:** Escribe **Web Server AMI**
 - **Descripción de la imagen:** Escribe **Lab AMI for Web Server**
6. Haz clic en **Crear imagen**.

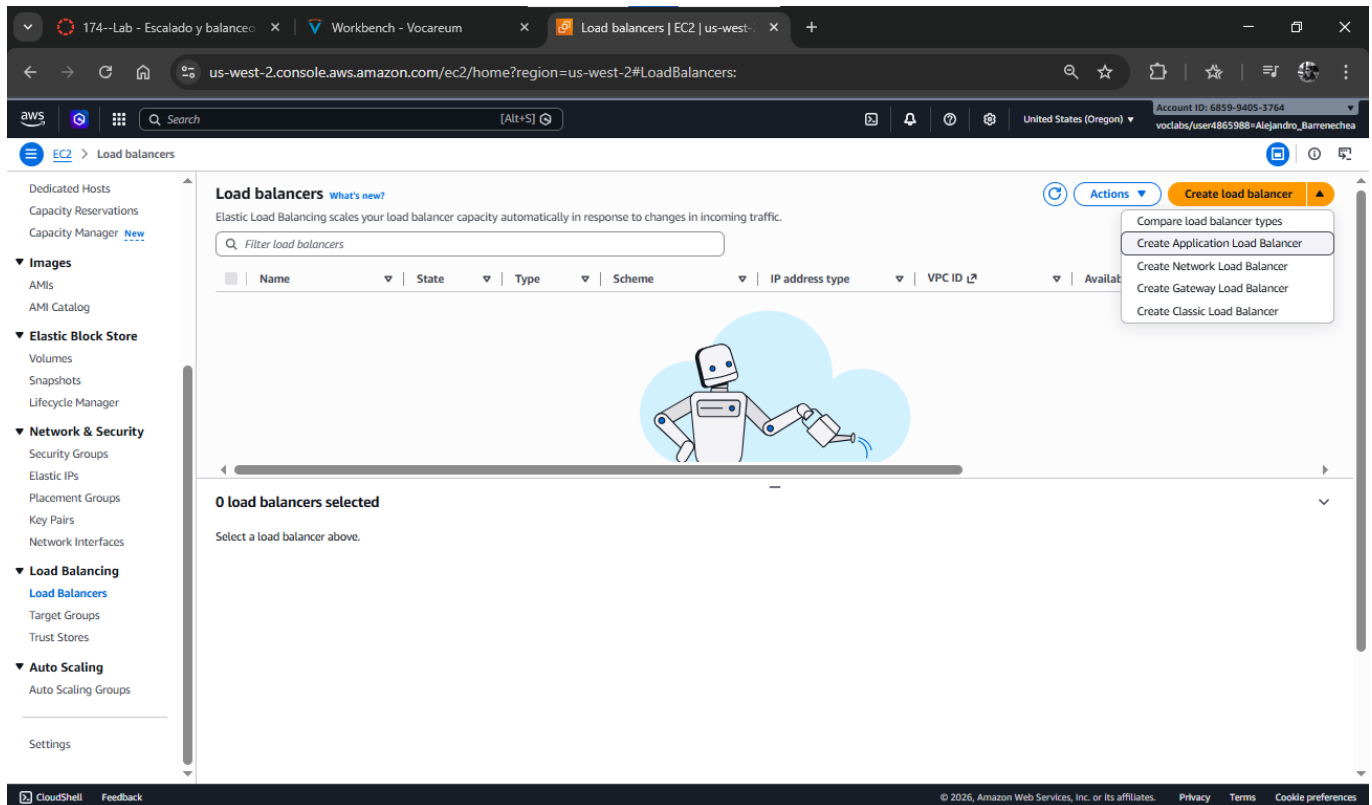
The screenshot shows the 'Create image' page in the AWS Management Console. The page title is 'Create image' and it includes a brief description: 'An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.' The 'Image details' section shows the 'Instance ID' as 'i-0d66ec9e49ac0c823 (Web Server 1)'. The 'Image name' field is set to 'Web Server AMI'. The 'Image description - optional' field is set to 'Lab AMI for Web Server'. The 'Reboot instance' checkbox is checked. The 'Instance volumes' section shows a table with columns: Storage type, Device, Snapshot, Size, Volume type, IOPS, Throughput, Delete on termination, and Encrypted. The table contains one row with 'EBS' as the storage type, '/dev/x...' as the device, 'Create new snapshot from v...' as the snapshot, '8' as the size, 'EBS General Purpose SSD - ...' as the volume type, '100' as the IOPS, and 'Enable' for both 'Delete on termination' and 'Encrypted'. There is an 'Add volume' button below the table. A note states: 'During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.' The page also includes a 'Tags - optional' section with a note: 'A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.'

Nota analítica: La consola te notificará el ID de la AMI creada. Has creado una "plantilla" de oro (Golden Image) la cual será reutilizada para que cada instancia lanzada por el Auto Scaling sea un clon idéntico del entorno original.

Tarea 2: Creación de un Balanceador de Carga

Crea un balanceador de carga (Application Load Balancer) para enrutar eficientemente el tráfico TCP/HTTP hacia las múltiples instancias EC2 a lo largo de diversas AZs.

1.  En el panel de navegación izquierdo, dirígete a la sección **Balanceo de carga** y selecciona **Balanceadores de carga**.
2.  Haz clic en **Crear balanceador de carga**.
3.  En la tarjeta de **Application Load Balancer**, haz clic en **Crear**.



4.  En **Configuración básica**:
 - **Nombre del balanceador de carga**: Escribe **LabELB**
5.  En la sección de **Mapeo de red**:
 - **VPC** (Opcional según interfaz de consola actual): Selecciona **Lab VPC**.
 -  Marca **ambas** zonas de disponibilidad listadas.
 -  Para la primera zona, selecciona **Public Subnet 1**.
 -  Para la segunda zona, selecciona **Public Subnet 2**.

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
Label

Scheme | info
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**
• Serves internet-facing traffic.
• Has public IP addresses.
• DNS name resolves to public IPs.
• Requires a public subnet.

☐ **Internal**
• Serves internal traffic.
• Has private IP addresses.
• DNS name resolves to private IPs.
• Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type | info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **Internet-facing** load balancers only.

Network mapping | info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-059x473d448ba9afe (Lab VPC)
10.0.0.0/16

IP pools | info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view [Pools](#) in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-west-2a (usw2-az1)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-08a510715371ec71 (Public Subnet 1)
10.0.0.0/24

☒ **us-west-2b (usw2-az2)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-080a86c4df3c5e566 (Public Subnet 2)
10.0.2.0/24

6. 🛡️ En la sección **Grupos de seguridad**:

- ✗ Elimina el grupo predeterminado haciendo clic en la "X".
- 🛡️ Selecciona del menú el grupo llamado **Web Security Group** (ya configurado para permitir HTTP).

7. 🎧 En **Agentes de escucha y enrutamiento**, haz clic en el enlace **Crear grupo de destino** (esto abrirá una pestaña nueva del navegador).

Security groups | info
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups

Web Security Group (sg-07a695c39b0be195)

Listeners and routing | info
A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP80 | Remove

Protocol
HTTP

Port
80
1-65535

Default action | info
The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action
☒ **Forward to target groups** ☐ Redirect to URL ☐ Return fixed response

Forward to target group | info
Choose a target group and specify routing weight or [create target group](#).

Target group
Select a target group

Weight
1
0-999

Percent
100%

+ Add target group
You can add up to 4 more target groups.

Target group stickiness | info
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add listener tag](#)
You can add up to 50 more tags.

[Add listener](#)
You can add up to 49 more listeners.

8. 📄 En la nueva pestaña, bajo **Configuración básica**:

- Tipo de destino:** Selecciona **Instancias**.
- Nombre del grupo de destino:** Escribe **lab-target-group**.
- ⬇️ Desplázate al fondo y haz clic en **Siguiente**.

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Select what resource type you want to target. Only the selected resource type can be registered to this target group.

- ☒ **Instance**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: **ALB** **NLB** **GWLB**
- ☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: **ALB** **NLB** **GWLB**
- ☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: **ALB**
- ☐ **Application Load Balancer**
Routes all of your IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: **NLB**

Target group name

Name must be unique per Region per AWS account.
lab-target-group
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters. Count: 16/32

Protocol

Protocol for communication between the load balancer and targets.
HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.
☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-059c473d448ba8afe (Lab VPC)

Protocol version

☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.
☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.
☐ **gRPC**

[Create VPC](#)

9. En la página de *Registrar destinos* (Register targets), no interactúes con la lista de instancias (el Auto Scaling se encargará luego) y simplemente haz clic en **Siguiente** (Next).

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1)

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-0d66ec9e49ac0a823	Web Server 1	Running	Web Security Group	us-west-2b	10.0.2.137	subnet-080a86c4dfb3cc566	April 3, 2026, 15:07 (UTC-08:00)

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.
80
1-65535 (separate multiple ports with comma)
[Include as pending below](#)

Review targets

Targets (0)

[Filter targets](#) ☒ Show only pending [Remove all pending](#)

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
No instances added yet								

Specify instances above, or leave the group empty if you prefer to add targets later.

0 pending [Cancel](#) [Previous](#) [Next](#)

10. En la página de *Revisar y crear* (Review and create), desliza hacia abajo y haz clic en **Crear grupo de destino**. Una vez creado, cierra la pestaña y vuelve a la original del Balanceador.

11. Haz clic en el ícono de **Actualizar** () junto a "Reenviar a" y elige tu grupo recién creado **lab-target-group**.

12. Presiona **Crear balanceador de carga** ubicado al final.
13. Entra en **Ver balanceador de carga** y copia el valor mostrado como **Nombre DNS** (guárdalo en tu portapapeles o en un anotador, es el endpoint principal).

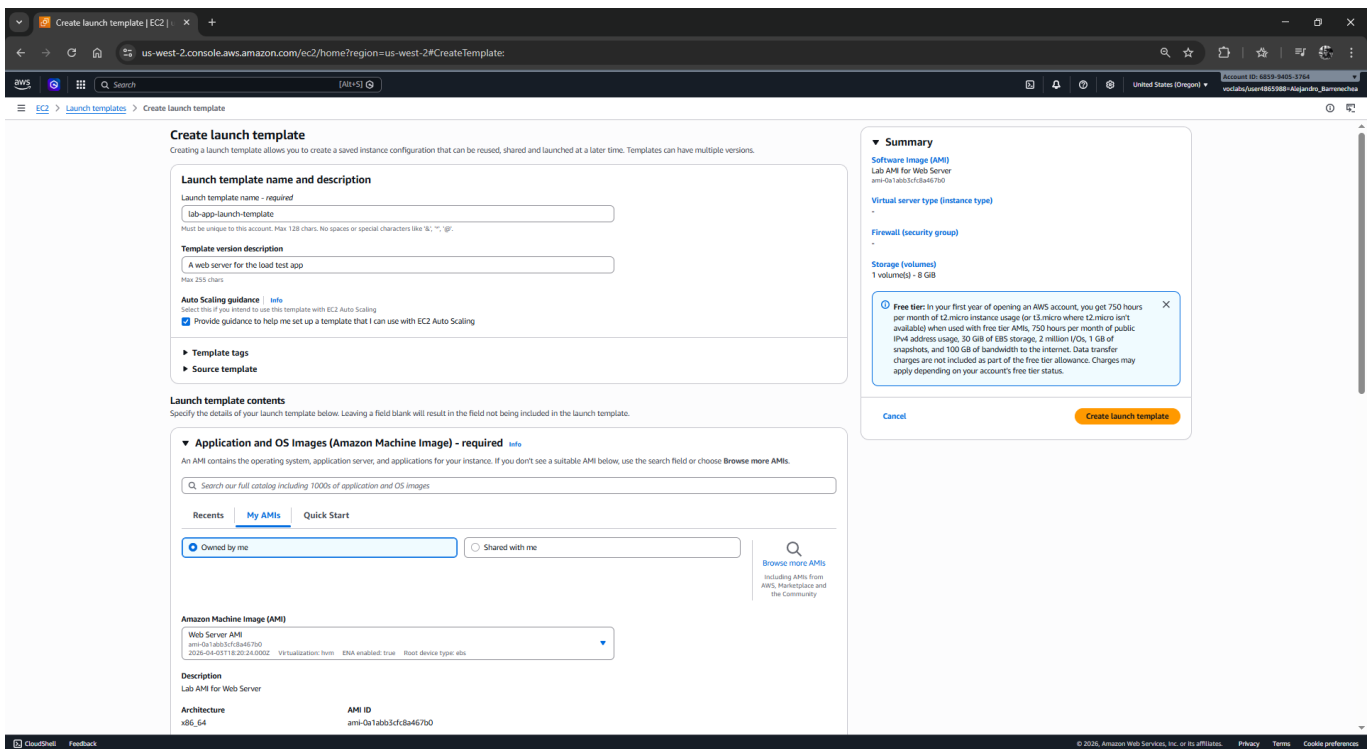
Tarea 3: Creación de una plantilla de lanzamiento (Launch Template)

Define una receta con los parámetros específicos (IAM, Tipo, Security Group, AMI) para la creación de las nuevas instancias gestionadas.

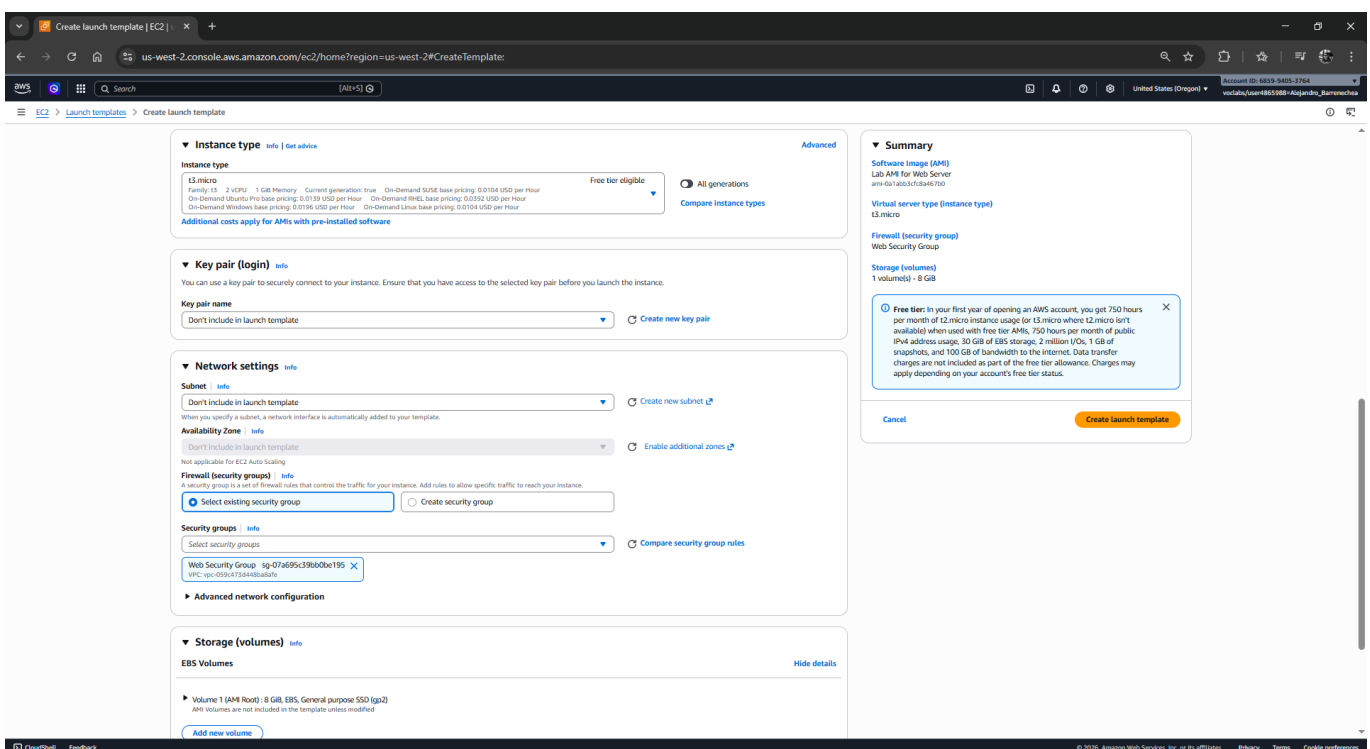
- Desde tu buscador superior accede nuevamente a la consola **EC2**.
- En el panel izquierdo, bajo **Instancias**, selecciona **Plantillas de lanzamiento**.
- Selecciona **Crear plantilla de lanzamiento**.

- En **Nombre y descripción de la plantilla de lanzamiento**:
 - Nombre de plantilla de lanzamiento:** Escribe **lab-app-launch-template**
 - Descripción de versión de la plantilla:** Escribe **A web server for the load test app**

- ☒ Marca la casilla: **Proporcionar orientación que me ayude a configurar...**
- 5.  En **Imágenes de aplicación y SO (AMI)**, posíciónate en la pestaña **Mis AMI** y verifica que **Web Server AMI** (la que creaste al inicio) se encuentre seleccionada automáticamente.



- 6.  En **Tipo de instancia**, selecciona **t3.micro** (o **t2.micro** si el primero da error debido a stock regional).
- 7.  En **Par de claves**, asegúrate de que esté configurado como **No incluir en la plantilla de lanzamiento**. (No conectaremos vía SSH).
- 8.  En **Configuraciones de red > Grupos de seguridad**, elige el **Web Security Group**.
- 9.  Haz clic en el botón naranja **Crear plantilla de lanzamiento** y luego en **Ver plantillas de lanzamiento**.



☑ Tarea 4: Creación de un Grupo de Auto Scaling

Utiliza la plantilla base creada para provisionar los recursos bajo capacidad controlada y segura.

1. ☑ Activa la casilla de tu **lab-app-launch-template**. Desde el menú desplegable **Acciones**, elige **Crear grupo de Auto Scaling**.

The screenshot shows the AWS Management Console interface for 'Launch Templates'. On the left, a navigation pane lists various services like EC2, IAM, and CloudWatch. The main area displays a table of launch templates. The first entry, 'lab-app-launch-template', is selected. A context menu is open over this entry, showing options like 'Launch instance from template', 'Modify template', and 'Create Auto Scaling group', which is highlighted.

Below the table, the details for the selected launch template are shown. It includes fields for 'Launch template ID', 'Launch template name', 'Default version', 'Created by', and 'Owner'. The 'Launch template version details' section shows the version number, description, date created, and other configuration details like AMI ID, instance type, and security groups.

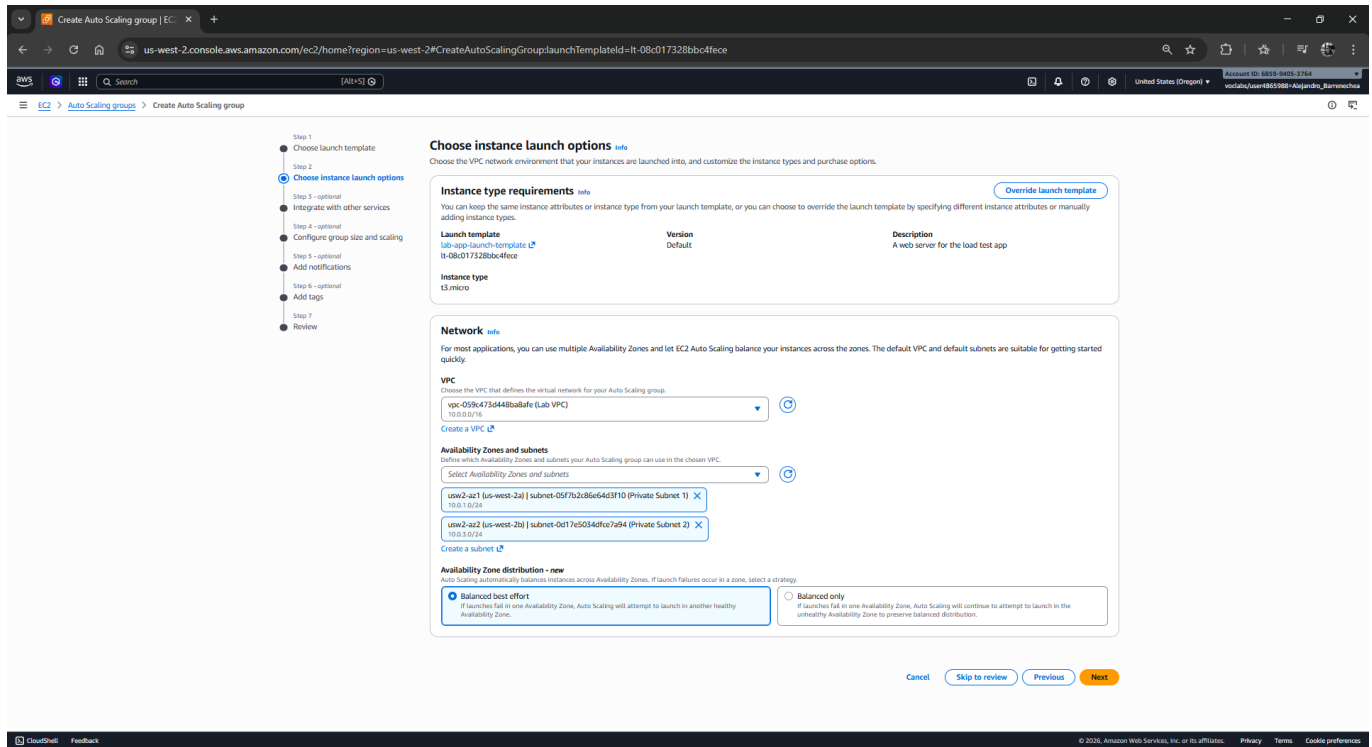
2. ✎ En la página de nombre, ingresa **Lab Auto Scaling Group** como **Nombre de grupo de Auto Scaling** y haz clic en **Siguiente**.

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console. The 'Choose launch template' step is active, and the 'lab-app-launch-template' is selected. The wizard prompts the user to enter a name for the Auto Scaling group, which is 'Lab Auto Scaling Group'. It also shows the launch template details, including the AMI ID, instance type, and security groups.

The 'Name' field is labeled 'Auto Scaling group name' and contains the text 'Lab Auto Scaling Group'. Below it, the 'Launch template' section shows the selected template and its details. The 'Version' is set to 'Default (1)'. The 'Description' is 'A web server for the load test app'. The 'Instance type' is 't3.micro'. The 'Request Spot Instances' option is set to 'No'.

3. 🏠 En la sección de red:
 - **VPC**: Elige **Lab VPC**.

- **Zonas de Disponibilidad y subredes:** Elige **Private Subnet 1 (10.0.1.0/24)** y **Private Subnet 2 (10.0.3.0/24)**. Esto garantiza que las instancias de computo finales no estén expuestas a internet directamente.
-  Haz clic en **Siguiente**.



4.  En opciones avanzadas de Balanceo de carga:
 -  Selecciona **Asociar a un balanceador de carga existente**.
 -  En opciones de grupo de destino, opta por **Elegir de los grupos de destino con balanceador de carga creados...**
 -  Selecciona **lab-target-group | HTTP**.
 -  En **Tipos de comprobación de estado**, marca la casilla de **ELB**.
 -  Haz clic en **Siguiente**.

Create Auto Scaling group | EC2

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#CreateAutoScalingGroup:launchTemplateId=lt-08c017328bb04f6ce

Auto Scaling group

Step 1: Choose launch template
Step 2: Choose instance launch options
Step 3: Integrate with other services
Step 4: optional - Configure group size and scaling
Step 5: optional - Add notifications
Step 6: optional - Add tags
Step 7: Review

Integrate with other services - optional

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ Attach to an existing load balancer
Choose from your existing load balancers.

☐ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach

☒ Choose from your load balancer target groups
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

lab-target-group | HTTP
Application Load Balancer: LABELB

VPC Lattice integration options

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

☒ No VPC Lattice service
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

Create new VPC Lattice service

Application Recovery Controller (ARC) zonal shift - new

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ Enable zonal shift
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

Always enabled

Create Auto Scaling group | EC2

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#CreateAutoScalingGroup:launchTemplateId=lt-08c017328bb04f6ce

Auto Scaling group

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

lab-target-group | HTTP
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Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

Always enabled

Additional health check types - optional

☒ Turn on Elastic Load Balancing health checks
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☒ EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing. To avoid unexpected terminations, first verify the settings of these health checks in the Load Balancer console

☐ Turn on VPC Lattice health checks
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300 seconds

Cancel Skip to review Previous Next

5. En la página de tamaño del grupo y las políticas:
- **Capacidad deseada:** 2
 - **Capacidad mínima:** 2
 - **Capacidad máxima:** 4
 - ☒ En **Políticas de escalado**, marca la opción **Política de escalado de seguimiento de destino** (Target tracking).
 - **Tipo de métrica:** Deja la opción en **Uso promedio de CPU**.
 - **Valor de destino:** Modifica a 50.
 - Haz clic en **Siguiente**.

Configure group size and scaling - optional

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and MemoryGB are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

2

Scaling

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

Equal or less than desired capacity

Max desired capacity

4

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy.

☐ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy

Choose a CloudWatch metric, and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Target Tracking Policy

Metric type

Monitored metrics that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

Target value

50

Instance warmup

300 seconds

☐ Disable scale in to create only a scale-out policy

Instance maintenance policy

6. Salta las opciones de las notificaciones haciendo clic en **Siguiente**.

7. En Etiquetas, haz clic en **Agregar etiqueta**:

- **Clave:** Name
- **Valor:** Lab Instance
- Haz clic en **Siguiente**.

Add tags - optional

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.

Tags (1)

Key	Value - optional	Tag new instances
Name	Lab Instance	☒

Add tag

4/9 remaining

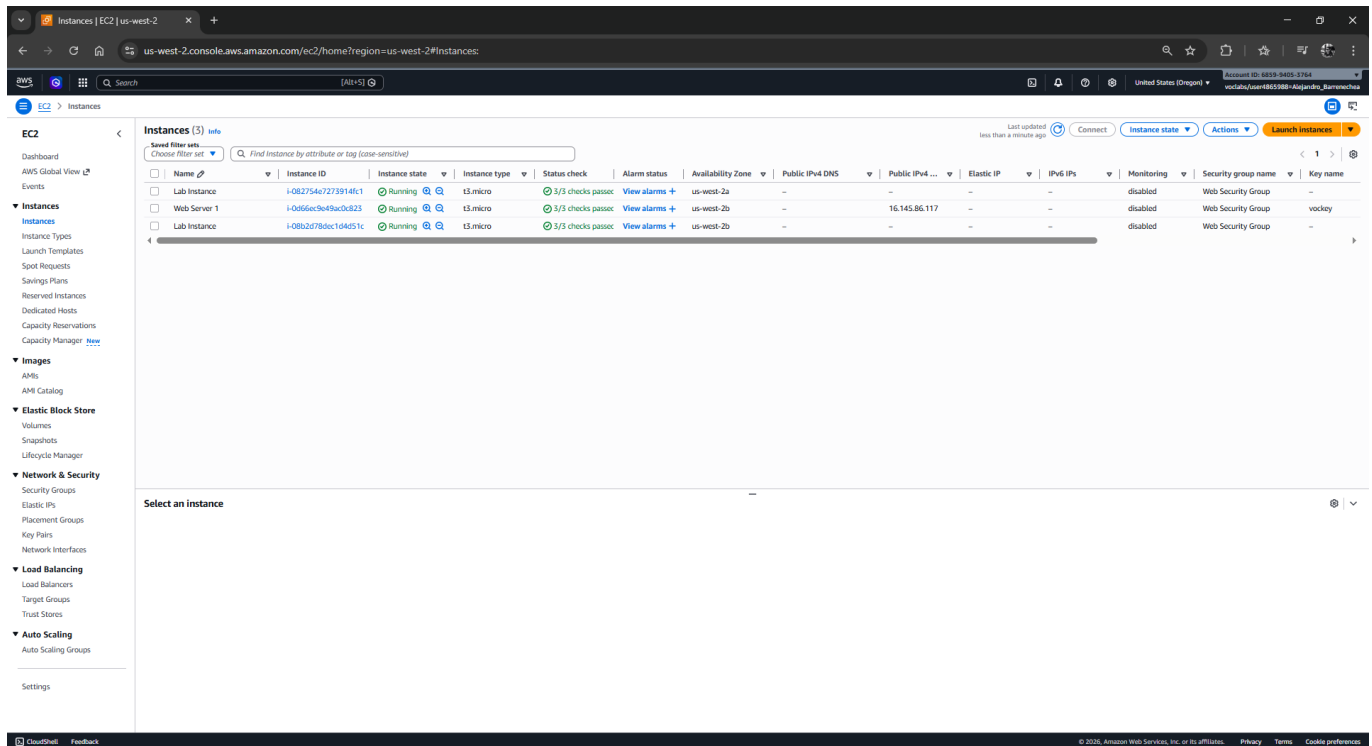
Cancel **Previous** **Next**

8. Revisa los datos y finaliza en **Crear grupo de Auto Scaling**. (Estas instancias tardarán un momento en lanzarse e iniciarse).

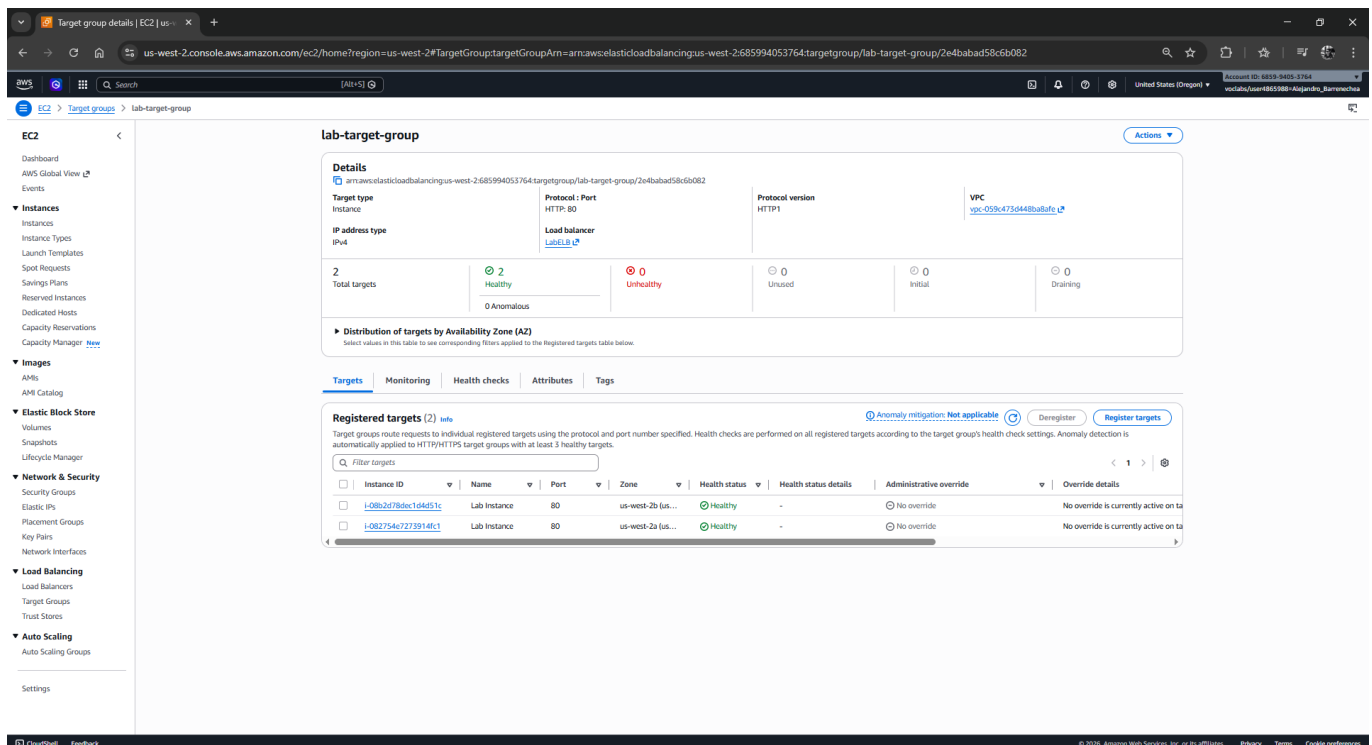
Tarea 5: Verificación del balanceo de carga

Comprueba que las instancias despliegan el servicio correctamente a través del grupo de destino bajo el paraguas del ALB.

1.  Sitúate nuevamente en el panel de navegación izquierdo en **Instancias** > **Instancias**. Verás hasta 2 instancias denominadas "Lab Instance" arrancando inicializadas por el grupo auto escalable.



2.  Navega al apartado de **Balanceo de Carga** > **Grupos de destino** y selecciona el tuyo (**lab-target-group**).
3.  En la pestaña **Destinos**, monitorea el **Estado de comprobación de estado** de ambas instancias "Lab Instance". (Presiona actualizar hasta que su estado cambie a **Healthy** / Saludable).

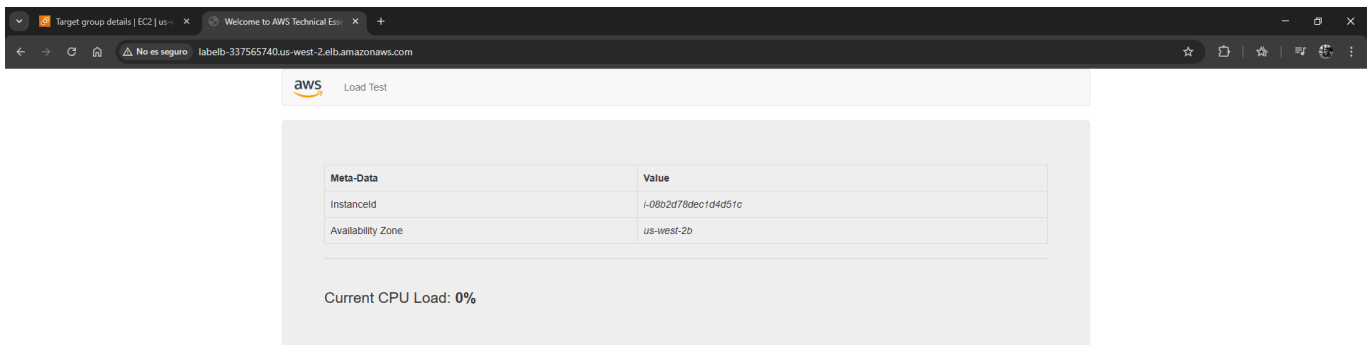


4. 🌐 Abre una nueva pestaña de navegador web, e ingresa el **Nombre DNS del balanceador de carga** (aquel que guardaste en el bloc de notas).
5. 🐞 Observarás tu aplicación llamada "Load Test" ejecutándose por el puerto HTTP. Felicidades, el balanceador funciona.

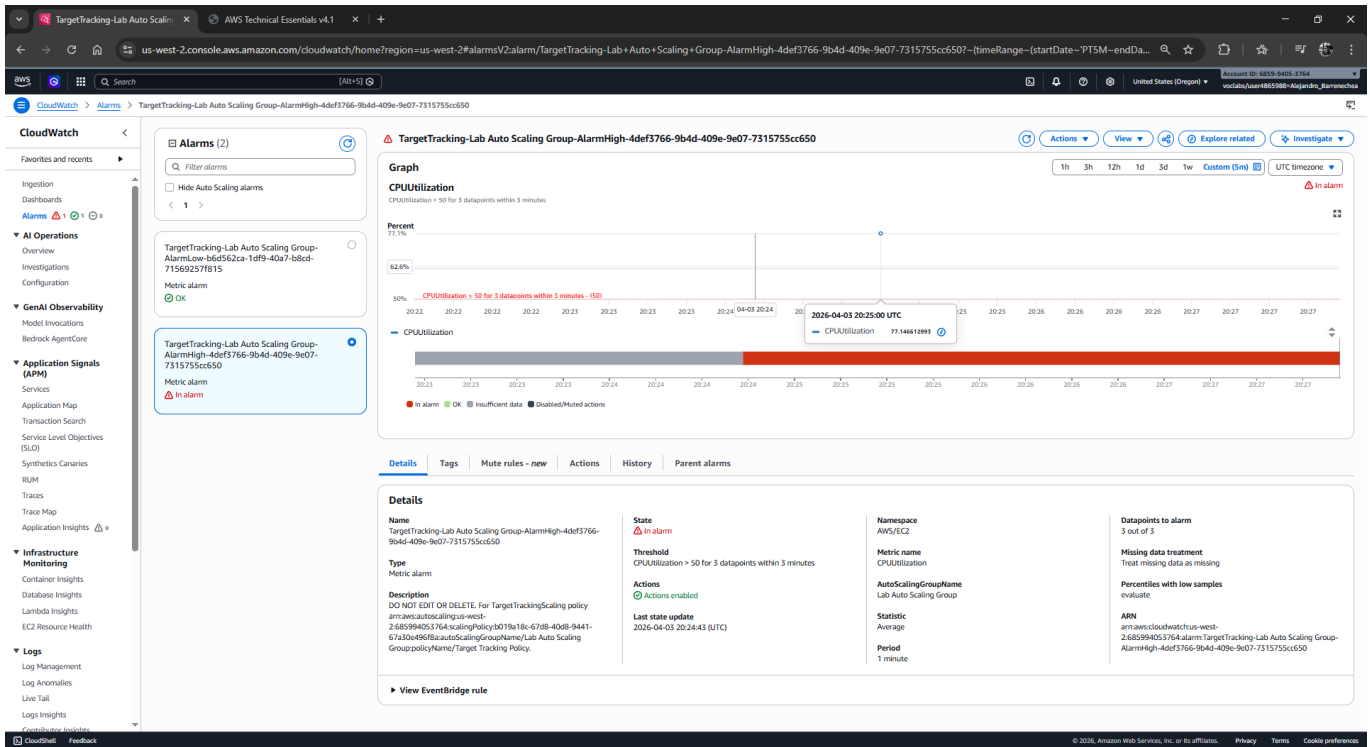
🛠️ Tarea 6: Prueba de la Elasticidad y Auto Scaling

Como configuraste la capacidad mínima en 2 instancias, hasta el momento los niveles están estables. Deberás generar un consumo masivo forzado para activar la recolección de métricas.

1. ☁ Regresa a tu consola AWS, abre la búsqueda e ingresa **CloudWatch** para abrir la consola del servicio de monitorización.
2. 🔔 En el panel izquierdo dirígete a **Alarmas > Todas las alarmas**. Allí encontrarás las alarmas generadas automáticamente por tu Auto Scaling (sus nombres actualmente inician con **TargetTracking-** seguido del nombre de tu grupo e incluyen términos como **AlarmHigh** o **AlarmLow**). *Si no las visualizas, verifica estar en la misma Región o búscalas desde EC2 > Grupos de Auto Scaling > tu grupo > pestaña "Escalado Automático".*
3. ✅ Identifica la alarma que detecta la subida de CPU (la que dice "High"). Su estado debe ser **OK** (Significa que por ahora los niveles de uso están bajo métricas normales y aceptables).
4. 🌐 Vuelve a la pestaña exterior de la aplicación web (La del balanceador de carga *Load Test app*) alojada en el navegador externo.
5. ⚡ Haz clic en el botón superior verde **Load Test**. (Esto saturará artificialmente el CPU al 100%).



6. 🔄 Regresa a **CloudWatch** y actualiza el gráfico. En el transcurso de aproximadamente 3 a 5 minutos, la alarma *AlarmHigh* pasará a encontrarse en estado de **En Alarma (In Alarm)** sobrepasando la línea de seguridad del 50%.



7. ☒ Por último, ve a la consola de **EC2 > Instancias**. Comprobarás que se han provisto y están arrancando automáticamente nuevas "Lab Instance" para mitigar la saturación de procesos respondiendo proactivamente al aumento de carga (Scaled Out).

The screenshot shows the AWS Management Console 'Instances' page. The table lists five instances:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name
Lab Instance	i-008cd1360e096c52	Running	t3.micro	3/3 checks passed	View alarms +	us-west-2a	-	-	-	-	disabled	Web Security Group	-
Lab Instance	i-082754c7273914f1	Running	t3.micro	3/3 checks passed	View alarms +	us-west-2a	-	-	-	-	disabled	Web Security Group	-
Lab Instance	i-01a52acbf1f6041a	Running	t3.micro	3/3 checks passed	View alarms +	us-west-2b	-	-	-	-	disabled	Web Security Group	-
Lab Instance	i-08b2d78dcf1d4d51c	Running	t3.micro	3/3 checks passed	View alarms +	us-west-2b	-	-	-	-	disabled	Web Security Group	-
Web Server 1	i-0a56ec3e49acdc23	Running	t3.micro	3/3 checks passed	View alarms +	us-west-2b	-	16.145.86.117	-	-	disabled	Web Security Group	vockey

Tarea 7: Terminación de la instancia original Web Server 1

1. Selecciona exclusivamente tu instancia semilla **Web Server 1**.
2. Entra en el menú **Estado de la instancia**, y selecciona **Terminar instancia** (Terminate).
3. Confirma el cuadro de diálogo. Esta instancia, habiendo derivado en la creación de la AMI inicial, ya no es necesaria dado que toda la infraestructura se apoya y sostiene en instancias abstraídas de la conexión directa detrás del Balanceador.

EC2

Instances

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Settings

us-west-2

Instances

Successfully initiated termination (deletion) of i-0d66ec9e49ac0c823

Instances (5)

Save filter sets

Choose filter set

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name
Lab Instance	i-008cd1360e0d9c52	Running	t3.micro	3/3 checks passed	View alarms	us-west-2a	-	-	-	-	disabled	Web Security Group	-
Lab Instance	i-08275467273914f1	Running	t3.micro	3/3 checks passed	View alarms	us-west-2a	-	-	-	-	disabled	Web Security Group	-
Lab Instance	i-07a33ac0f9f6041a	Running	t3.micro	3/3 checks passed	View alarms	us-west-2b	-	-	-	-	disabled	Web Security Group	-
Lab Instance	i-08b2a780ec104d51c	Running	t3.micro	3/3 checks passed	View alarms	us-west-2b	-	-	-	-	disabled	Web Security Group	-
Web Server 1	i-0d66ec9e49ac0c823	Terminated	t3.micro	-	View alarms	us-west-2b	-	-	-	-	disabled	-	vockey

Select an instance

CloudShell

Feedback

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✓ ¡Felicidades! Has concluido este laboratorio.